COER UNIVERSITY

Lab Sheet 2: Data Manipulation Language (DML) Commands

Question 1 Question: Insert one record into the Student table.

Answer:

Explanation: Inserts a single row into the Student table specifying explicit values for all columns.

Question 2 Question: Insert five student records using separate INSERT statements.

Answer:

Explanation: Executes five individual INSERT statements to populate five distinct records into the table.

Question 3 Question: Insert multiple records into the Student table using a single statement.

Answer:

Explanation: Combines multiple value tuples in a single INSERT statement for multi-row insertion.

Question 4 Question: Insert records into the Employee table.

Answer:

Explanation: Populates sample employee records into the Employee table.

Question 5 Question: Insert records into the Department table.

Answer:

Explanation: Populates department records into the Department table.

Question 6 Question: Insert records into the Product table.

Answer:

Explanation: Populates product records into the Product table.

Question 7 Question: Insert records into the Customer table.

Answer:

Explanation: Populates initial records into the Customer table.

Question 8 Question: Insert records into the Course table.

Answer:

Explanation: Inserts course details into the Course table.

Question 9 Question: Insert a student record with values for all columns.

Answer:

Explanation: When inserting values for every column in exact schema order, explicit column names can be omitted.

Question 10 Question: Insert a student record by specifying only selected columns.

Answer:

Explanation: Specifies only mandatory columns (Student_ID, Name, Course), while unmentioned columns (City, Fees) take NULL.

Question 11 Question: Update the city of a particular student.

Answer:

Explanation: Uses a WHERE clause to update the city specifically for Student_ID = 101.

Question 12 Question: Update the salary of a particular employee.

Answer:

Explanation: Modifies the salary for the employee with EmpID = 1.

Question 13 Question: Increase the salary of all employees by 10%.

Answer:

Explanation: Omitting the WHERE clause applies a 10% raise across all records in the Employee table.

Question 14 Question: Increase the price of all products by 5%.

Answer:

Explanation: Applies a 5% price increase to all products in the Product table.

Question 15 Question: Change the course of a particular student.

Answer:

Explanation: Updates the course for Student_ID = 102.

Question 16 Question: Update the department of an employee.

Answer:

Explanation: Changes the department attribute for EmpID = 2.

Question 17 Question: Update the city of all students from Meerut to Delhi.

Answer:

Explanation: Updates City to 'Delhi' for all records where City currently equals 'Meerut'.

Question 18 Question: Update the fees of a particular course.

Answer:

Explanation: Modifies the course fees for CourseID = 301.

Question 19 Question: Update multiple columns of a student simultaneously.

Answer:

Explanation: Updates Course, City, and Fees simultaneously in a single UPDATE statement.

Question 20 Question: Delete a particular student record using Student_ID.

Answer:

Explanation: Removes the student record corresponding to Student_ID = 110.

Question 21 Question: Delete all students belonging to a particular city.

Answer:

Explanation: Deletes all rows where City equals 'Meerut'.

Question 22 Question: Delete employees whose salary is below a specified amount.

Answer:

Explanation: Deletes employee records earning less than 40,000.

Question 23 Question: Delete products whose quantity is zero.

Answer:

Explanation: Deletes out-of-stock products (Quantity = 0).

Question 24 Question: Delete records satisfying multiple conditions using AND/OR.

Answer:

Explanation: Deletes student records that satisfy both conditions simultaneously.

Question 25 Question: Perform a complete sequence of INSERT, UPDATE and DELETE operations on a Student table and display the table after each operation.

Answer:

Explanation: Demonstrates a full CRUD sequence (Create, Read, Update, Delete) on a single record, checking table state after each step.